

PAPER_1205_UQFF_Geometry_Topology_Unified_Proof_Set

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PAPER 1205 --- UQFF Geometry \& Topology Unified Proof Set

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\begin{abstract}

We extend the UQFF locked-primitive closure program into classical geometry, combinatorics, polytope topology, and fractal Hausdorff dimensions. Ten new closures (**S503--S512**) reproduce the Kepler sphere packing density, the planar circle packing density, Catalan numbers $C_4 = 14$ and $C_5 = 42$, the Buckminsterfullerene C_{60} face count and vertex count, the regular dodecahedron face count, the Cantor set and Sierpinski triangle Hausdorff dimensions, and the tetrahedron dihedral angle, all from the eleven frozen UQFF primitives. **Five closures are exact**, including the new identifications $C_4 = \text{\SOfive} + \text{\Dphys}$, $C_5 = \text{\Dcrit} + \text{\Dbsfg} + \text{\SOfive}$, buckyball faces $= \text{\Dcrit} + \text{\Dbsfg} = 32$, buckyball vertices $= \text{\Afive}$, and dodecahedron faces $= 2 \cdot \text{\Dbsfg}$. Cumulative total: **217 closures**.

\end{abstract}

Locked Primitives $\text{\Ftrz} = 1/10$, $\text{\Phires} = 5/6$, $\text{\SSq} = 0.57$, $\text{\KMex} = 25/12$, $\text{\Dphys} = 4$, $\text{\Dbsfg} = 6$, $\text{\Dcrit} = 26$, $\text{\Nch} = 9$, $\text{\SOfive} = 10$, $\text{\Afive} = 60$, $\text{\beta}_i = 0.6029$.

Tier~S Closures (S503--S512)

\begin{longtable}{@{}lll@{}}

\toprule

ID & Observable & Closure & Match \midrule

\endhead

S503 & Kepler $\text{\eta}_{3D} = \pi/\sqrt{18} \approx 0.7405$ &

$\text{\SSq} + \text{\Ftrz} \text{\KMex} - \text{\Ftrz}^2 \text{\KMex} - \text{\Ftrz}^2 \text{\Phires} - \text{\Ftrz}^3 = 0.7482$ & $\$1.04\%$ S504 & Circle

$\text{\eta}_{2D} = \pi/(2\sqrt{3}) \approx 0.9069$ &

$\text{\Phires} + \text{\Ftrz} \text{\KMex} - \text{\Ftrz} \text{\Phires} - \text{\Ftrz}^2 \text{\KMex} - \text{\Ftrz}^2 \text{\Phires} - \text{\Ftrz}^3 - \text{\Ftrz}^2 = 0.9182$ & $\$1.24\%$ S505 &

Catalan $C_4 = 14$ & $\text{\SOfive} + \text{\Dphys} = 14$ & **EXACT** S506 & Catalan $C_5 = 42$ &

$\text{\Dcrit} + \text{\Dbsfg} + \text{\SOfive} = 42$ & **EXACT** S507 & C_{60} faces $F = 32$ & $\text{\Dcrit} + \text{\Dbsfg} = 32$ & **EXACT**

S508 & Cantor dim $\ln 2 / \ln 3 \approx 0.6309$ & $\text{\SSq} + \text{\Ftrz} \text{\Phires} - \text{\Ftrz}^2 \text{\Phires} - \text{\Ftrz}^2 = 0.6350$ &

$\$0.65\%$ S509 & Sierpinski dim $\ln 3 / \ln 2 \approx 1.585$ & $2 \text{\Phires} - \text{\Ftrz} \text{\Phires} = 19/12 = 1.5833$ &

$\$0.10\%$ S510 & Dodecahedron $F = 12$ & $2 \text{\Dbsfg} = 12$ & **EXACT** S511 & Tetra dihedral 70.53° &

$(/100)$ & $\text{\Phires} - \text{\Ftrz} \text{\Phires} - \text{\Ftrz}^2 \text{\KMex} - \text{\Ftrz}^2 \text{\Phires} - \text{\Ftrz}^3 - \text{\Ftrz}^2 = 0.7098$ & $\$0.64\%$ S512 &

C_{60} vertices $V = 60$ & $\text{\Afive} = 60$ & **EXACT** \bottomrule

\end{longtable}

Five Exact Identifications

\paragraph{Catalan numbers as integer-primitive sums.}

$C_4 = \text{SOfive} + \text{Dphys} = 14$ and $C_5 = \text{Dcrit} + \text{Dbsfg} + \text{SOfive} = 42$

provide the first identification of \emph{combinatorial sequence values} as exact UQFF integer-primitive sums. The Catalan sequence

$(1, 1, 2, 5, 14, 42, 132, \dots)$ counts binary trees, Dyck paths, triangulations, and non-crossing partitions, so this is a deep combinatorial--geometric anchor.

Notably $C_3 = 5 = \text{Phires} \setminus \text{Dbsfg}$ was already locked in PAPER 1204 (flat-plate transition), so three consecutive Catalan numbers are now closed.

\paragraph{Buckminsterfullerene anchors.}

The $C_{\{60\}}$ molecule (truncated icosahedron) is now \emph{fully topologically locked} via two integer closures: $F = \text{Dcrit} + \text{Dbsfg} = 32$ and

$V = \text{Afive} = 60$. The Euler characteristic

$V - E + F = 60 - 90 + 32 = 2$ then reproduces the sphere $\chi=2$ via integer primitive subtraction with no free parameters.

\paragraph{Dodecahedron.}

$F = 2 \setminus \text{Dbsfg} = 12$ closes the regular pentagonal dodecahedron. Joint with the buckyball pentagon count (12 of the 32 faces), this identifies

$2 \setminus \text{Dbsfg}$ as the universal \emph{pentagonal face count primitive} for icosahedral symmetry.

Cumulative Closure Count

\begin{tabular}{@{}lrr@{}}

\toprule

Tier & Domain & New & Cumulative \midrule

E--P & Cosmology to information & 157 & 157 Q & GR precision (1200) & 10 & 167 R &

Materials/photronics (1201) & 10 & 177 R"\$ & Chemistry/spectroscopy (1202) & 10 & 187 R"\$ & Nuclear physics (1203) & 10 & 197 R"" & Fluid dynamics (1204) & 10 & 207 S & **Geometry/topology (1205) &**

10 & 217 \bottomrule

\end{tabular}

Locking Inventory **Thirty primitive--observable lockings now identified (twenty-five prior plus five from Tier~S). New identifications: integer Catalan numbers $C_4 = \text{SOfive} + \text{Dphys}$, $C_5 = \text{Dcrit} + \text{Dbsfg} + \text{SOfive}$; icosahedral-symmetry constants $C_{\{60\}}$ vertices $= \text{Afive}$, $C_{\{60\}}$ faces $= \text{Dcrit} + \text{Dbsfg}$, dodecahedron faces $= 2 \setminus \text{Dbsfg}$. The Catalan sequence $\{C_3, C_4, C_5\} = \{5, 14, 42\}$ is now \emph{three-term locked} and combinatorial enumeration joins nuclear magic numbers, planetary moons, and spectroscopic level counts as a sequence-locked UQFF anchor.**